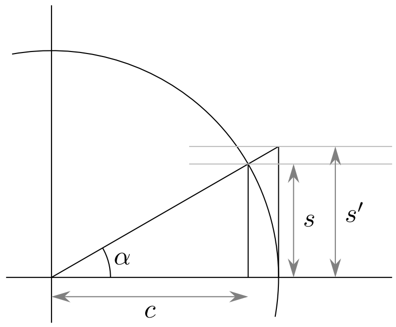
Derive the derivative of $\sin x$ from first principles.

Consider the angle α and the unit circle so that α is the length of that portion of the circumference that the angle subtends.



There is a small right triangle with base c that lies entirely within the α-section of the circle that in turn lies within the larger right triangle with base 1. Thus considering the areas we have

$$\frac{1}{2}sc \leq \frac{\alpha}{2\pi}\pi 1^2 \leq \frac{1}{2}s'1$$

or

$$sc \leq \alpha \leq s'.$$

Since $s = \sin\alpha$, $c = \cos\alpha$ and $s' = \tan\alpha$ the first and second inequalities above imply that

$$\frac{\sin\alpha}{\alpha} \leq \frac{1}{\cos\alpha}$$

and

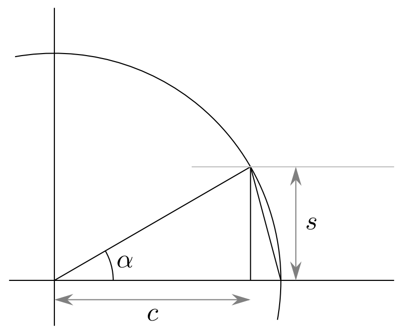
$$\cos\alpha \leq \frac{\sin\alpha}{\alpha}$$

respectively. Putting these together yields

$$\cos\alpha \leq \frac{\sin\alpha}{\alpha} \leq \frac{1}{\cos\alpha}$$

from which it is clear that

$$\lim_{\alpha\to 0}\frac{\sin\alpha}{\alpha} = 1. \tag{1}$$



Now draw the chord of the circle that also serves as the hypotenuse of the right triangle whose base has length $1 - c$. The squared length of this chord is less than the squared length of the arc so by Pythagoras

$$s^2 + (1 - c)^2 \leq \alpha^2$$

or

$$0 \leq (1 - c)^2 \leq \alpha^2 - s^2$$

and

$$0 \leq \left(\frac{1 - \cos\alpha}{\alpha}\right)^2 \leq 1 - \left(\frac{\sin\alpha}{\alpha}\right)^2.$$

This and equation 1 imply that

$$\lim_{\alpha\to 0} \frac{1 - \cos\alpha}{\alpha} = 0. \tag{2}$$

Finally, the derivative of $\sin x$ is the limit as α goes to zero of

$$\begin{aligned} \frac{\sin(x + \alpha) - \sin x}{\alpha} &= \frac{\sin x \cos\alpha + \cos x \sin\alpha - \sin x}{\alpha} \\ &= \sin x \frac{(\cos\alpha - 1)}{\alpha} + \cos x \frac{\sin\alpha}{\alpha} \end{aligned}$$

and equations 1 and 2 make it evident that this expression goes to $\cos x$.